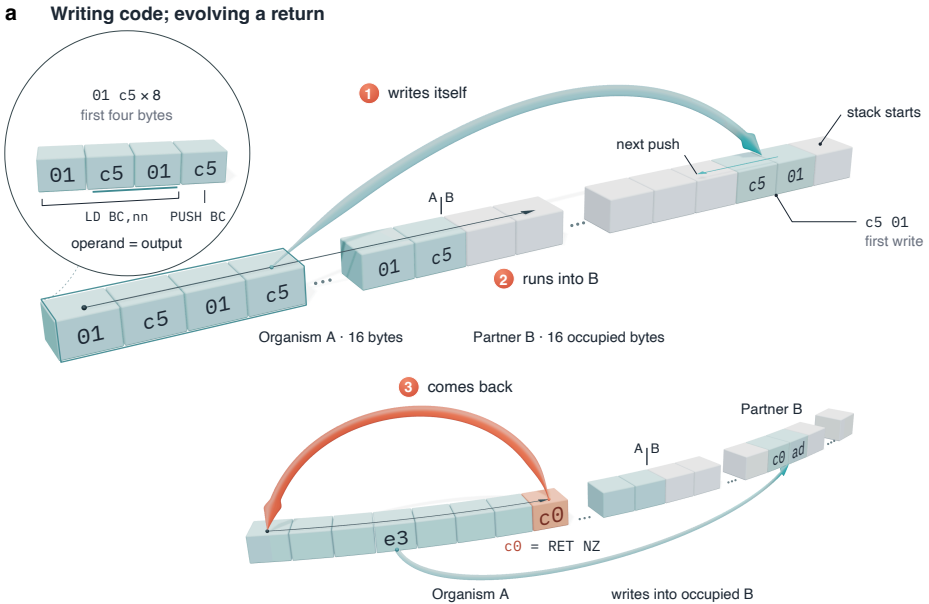
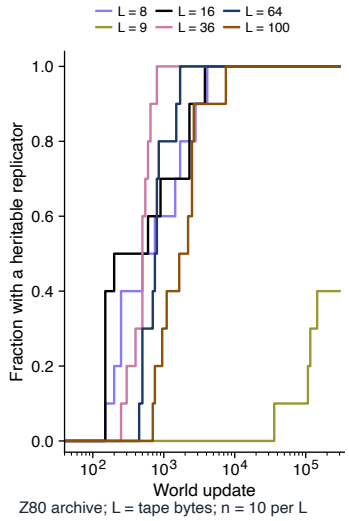
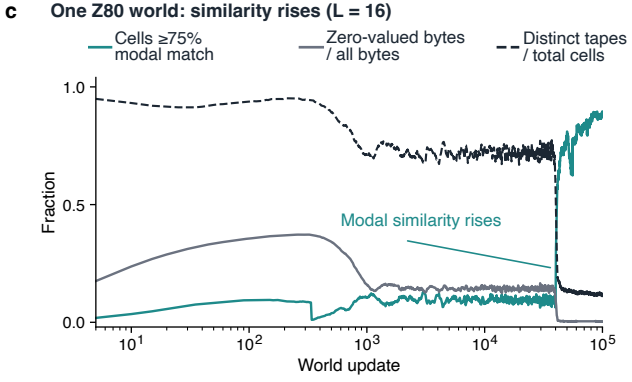
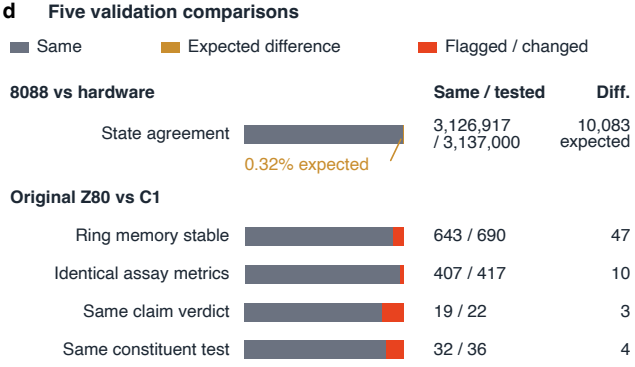

a
Writing code; evolving a return
01 c5 × 8
first four bytes
01 c5 01 c5
LD BC,nn
PUSH BC
operand = output
1 writes itself
stack starts
next push
c5 01
first write
A | B
2 runs into B
Organism A · 16 bytes
Partner B · 16 occupied bytes
3 comes back
Partner B
e3 c0
c0 = RET NZ
Organism A
writes into occupied B


b
Emergence from random code
L = 8
L = 16
L = 64
L = 9
L = 36
L = 100
Fraction with a heritable replicator
World update
Z80 archive; L = tape bytes; n = 10 per L


c
One Z80 world: similarity rises (L = 16)
Cells ≥75% modal match
Zero-valued bytes / all bytes
Distinct tapes / total cells
Fraction
Modal similarity rises
World update
Seed 2002; match to the contemporaneous modal tape


d
Five validation comparisons
Same
Expected difference
Flagged / changed
8088 vs hardware
Same / tested
Diff.
State agreement
3,126,917 / 3,137,000
10,083 expected
0.32% expected
Original Z80 vs C1
Ring memory stable
643 / 690
47
Identical assay metrics
407 / 417
10
Same claim verdict
19 / 22
3
Same constituent test
32 / 36
4
C1 = corrected Z80 instruction semantics